PROJECT REPORT

1. INTRODUCTION

1.1 Project Overview

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Project: Heart Disease Analysis using Tableau

Heart disease is one of the leading causes of death worldwide and continues to pose significant challenges to healthcare systems. Factors such as unhealthy lifestyles, poor dietary habits, smoking, obesity, diabetes, and physical inactivity contribute to the increasing prevalence of cardiovascular diseases. Although healthcare organizations collect large volumes of patient data, extracting meaningful insights from this information remains a challenge due to its complexity and lack of effective visualization.

This project presents **Heart Disease Analysis using Tableau**, an interactive business intelligence solution designed to transform raw healthcare data into meaningful visual insights. The project analyzes patient demographics, lifestyle habits, and medical conditions to identify factors associated with heart disease. Using Tableau's interactive dashboards, KPI cards, filters, calculated fields, and data storytelling capabilities, users can explore healthcare trends, compare patient groups, and understand disease patterns through intuitive visualizations.

The project includes **three interactive dashboards**, **ten analytical visualizations**, **KPI cards**, **interactive filters**, and a **Tableau Story**, all integrated into a **Flask-based web application** for improved accessibility. The web application provides users with a centralized platform to explore dashboards, review analytical findings, and access project information through an organized and user-friendly interface.

The insights generated from this project can assist healthcare professionals, researchers, and students in understanding the relationship between demographic characteristics, lifestyle factors, and medical conditions associated with heart disease. By converting healthcare data into actionable visual information, the project promotes data-driven decision-making, enhances awareness of cardiovascular risk factors, and demonstrates the practical application of business intelligence in the healthcare domain.

Scenario 1: Dr. Sharma – Cardiologist

Dr. Sharma is a senior cardiologist working at a metropolitan hospital. She wants to understand which lifestyle factors contribute most to the increase in heart disease cases among middle-aged patients. Using Tableau dashboards, she can analyze patient data segmented by age, gender, BMI, cholesterol levels, and smoking habits. This helps her identify high-risk groups and design targeted awareness campaigns, such as advising patients on weight management and smoking cessation programs.

Scenario 2: Ramesh – Healthcare Administrator

Ramesh works with a government health department and is tasked with developing preventive health policies. He uses Tableau dashboards to study trends in heart disease prevalence across different regions, comparing rural and urban populations. By analyzing correlations between sedentary lifestyle indicators and disease rates, he can recommend policies such as fitness programs in workplaces, stricter tobacco regulations, and subsidies for healthier food options. Tableau helps him present these findings in interactive dashboards for decision-makers.

Scenario 3: Anita – Patient

Anita, a 45-year-old professional with a family history of heart disease, wants to monitor her health risks. With simplified Tableau dashboards provided by her healthcare provider, she can visualize her risk factors such as cholesterol levels, blood pressure, and lifestyle habits compared to healthy benchmarks. The dashboard highlights actionable steps like increasing physical activity or reducing fat intake. This empowers Anita to make informed decisions about her lifestyle and proactively reduce her risk of developing heart disease.

1.2 Purpose

The primary purpose of this project is to develop an interactive Tableau-based healthcare analytics platform that transforms patient data into meaningful visual insights for heart disease analysis. The project aims to simplify the exploration of healthcare information by presenting demographic, lifestyle, and clinical indicators through dashboards, KPI cards, interactive filters, and data storytelling.

The solution enables users to identify heart disease trends, compare patient groups, and understand relationships between risk factors such as age, BMI, smoking, diabetes, stroke history, and physical activity. By integrating Tableau with a Flask web application, the project provides an accessible platform for viewing analytical dashboards and stories through a web interface.

This project demonstrates the effective use of business intelligence and data visualization techniques in healthcare, supporting data-driven decision-making, promoting preventive healthcare awareness, and improving the interpretation of complex medical datasets.

2. IDEATION PHASE

The ideation phase focused on understanding the healthcare problem, identifying stakeholder needs, and exploring possible analytical solutions before the development of the Tableau dashboard. Various healthcare perspectives, patient requirements, and visualization approaches were considered to ensure the final solution effectively supports heart disease analysis and data-driven decision-making.

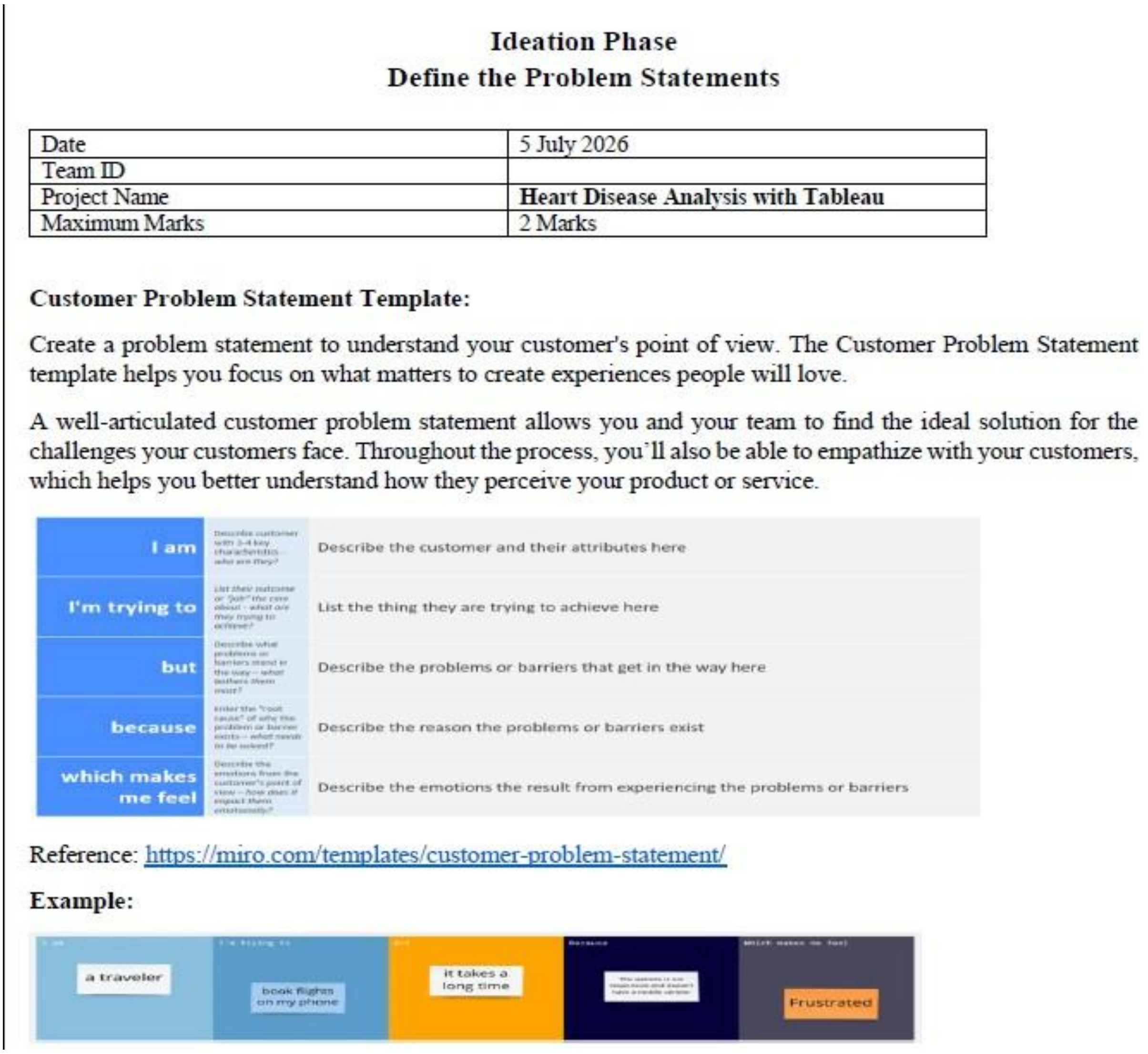
2.1 Problem Statement

Heart disease continues to be one of the leading causes of mortality worldwide, with its prevalence increasing due to factors such as unhealthy lifestyles, obesity, smoking, diabetes, and physical inactivity. Although healthcare organizations collect large volumes of patient information, extracting meaningful insights from this data remains challenging because raw datasets are difficult to interpret without effective visualization.

Traditional data analysis methods often require significant time and technical expertise, making it difficult for healthcare professionals and researchers to quickly identify disease patterns, high-risk patient groups, and relationships between demographic, lifestyle, and clinical factors. As a result, opportunities for early awareness, preventive healthcare, and informed decision-making may be limited.

This project addresses these challenges by developing an interactive **Heart Disease Analysis Dashboard** using Tableau. The solution transforms raw healthcare data into meaningful visual insights through interactive dashboards, KPI cards, filters, calculated fields, and Tableau Story. By integrating the analytical dashboards into a Flask-based web application, the project provides an accessible platform that enables users to explore healthcare trends, identify risk factors, and support data-driven healthcare analysis.

Figure 2.1: Problem Statement Canvas



Ideation Phase
Define the Problem Statements

Date	5 July 2026
Team ID	
Project Name	**Heart Disease Analysis with Tableau**
Maximum Marks	2 Marks

Customer Problem Statement Template:

Create a problem statement to understand your customer's point of view. The Customer Problem Statement template helps you focus on what matters to create experiences people will love.

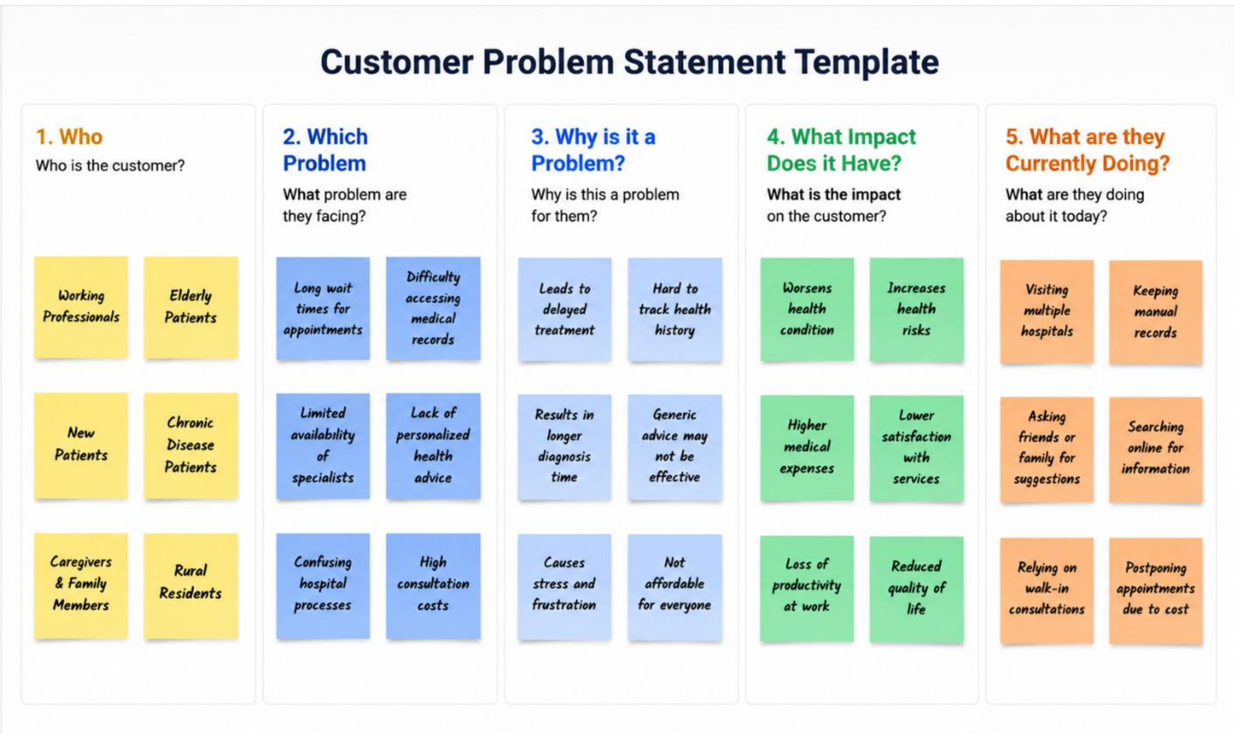
A well-articulated customer problem statement allows you and your team to find the ideal solution for the challenges your customers face. Throughout the process, you'll also be able to empathize with your customers, which helps you better understand how they perceive your product or service.

I am	Describe the customer and their attributes here
I'm trying to	List the thing they are trying to achieve here
but	Describe the problems or barriers that get in the way here
because	Describe the reason the problems or barriers exist
which makes me feel	Describe the emotions the result from experiencing the problems or barriers

Reference: https://miro.com/templates/customer-problem-statement/

Example:

a traveler	book flights on my phone	it takes a long time		Frustrated

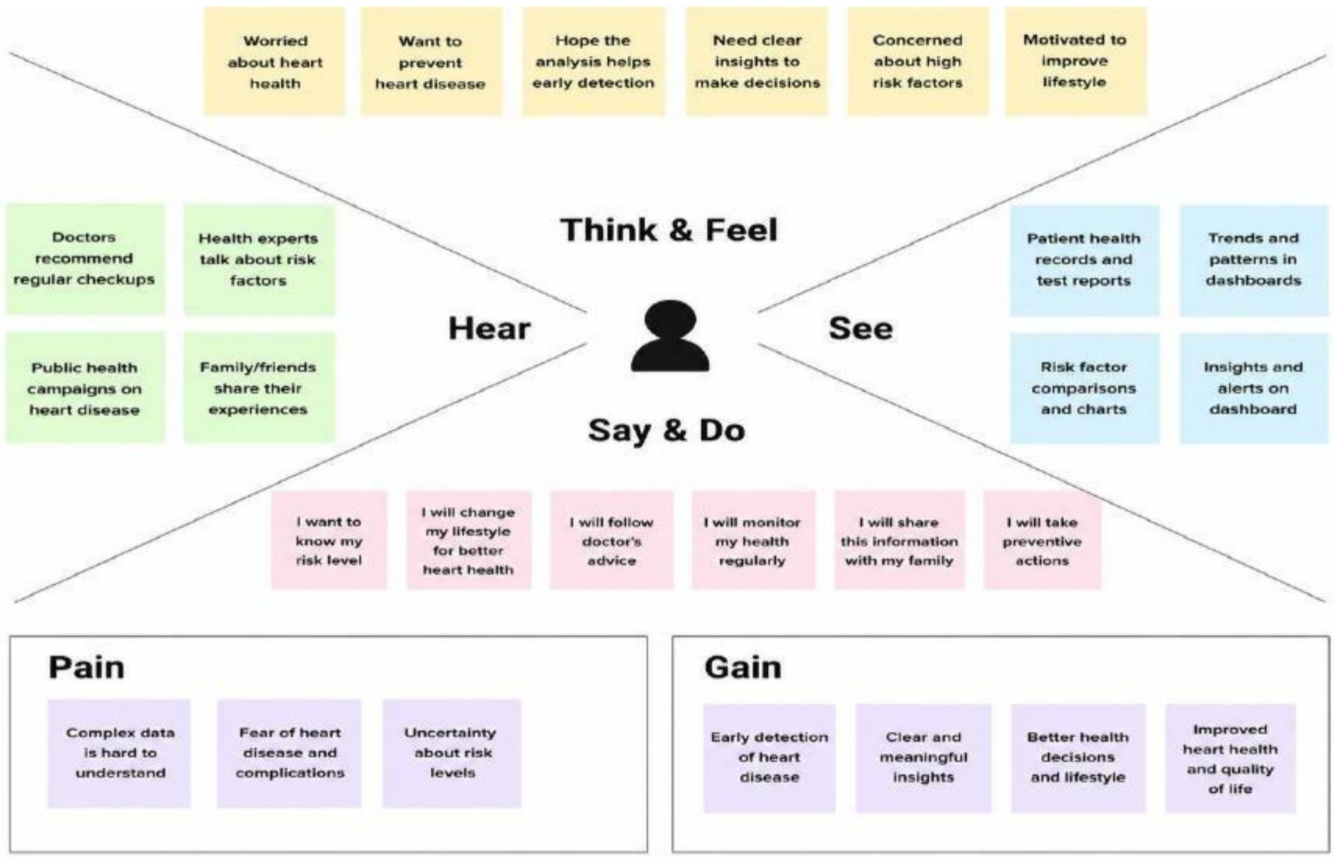


Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Hospital Administrator	Monitor patient statistics	Time-consuming reports	Manual data analysis	Inefficient in decision-making
PS-2	Patient	Understand personal health risks	Difficult medical information	Lack of visual explanations	Uncertain about health decisions

2.2 EMPATHY MAP

The Empathy Map Canvas was developed to understand the needs, expectations, challenges, and behaviors of the key stakeholders involved in heart disease analysis. It provides valuable insights into how different users interact with healthcare data and highlights the importance of presenting complex medical information through simple and interactive visualizations. Understanding these perspectives helped in designing a user-centric dashboard that supports efficient analysis and informed decision-making.

Figure 2.2: Empathy Map Canvas



The Empathy Map illustrates the perspectives of the primary stakeholders, including the cardiologist, healthcare administrator, and patient. It captures what they think, observe, say, and do while interacting with healthcare data, along with their challenges and expectations. These insights guided the development of interactive dashboards, KPI cards, filters, and Tableau Story to ensure the solution effectively supports healthcare analysis, risk identification, and data-driven decision-making.

2.2 EMPATHY MAP CANVAS

1. Dr. Sharma – Cardiologist

- **Says:** "Which patient groups are at the highest risk of heart disease? How can early intervention improve patient outcomes?"
- **Thinks:** Needs reliable insights to identify risk factors and support preventive treatment strategies.
- **Does:** Analyzes patient data using Tableau dashboards, comparing age, BMI, diabetes, smoking habits, and cholesterol-related indicators.
- **Feels:** Responsible for improving patient health through timely diagnosis and evidence-based clinical decisions.

Ideation Phase
Empathize & Discover

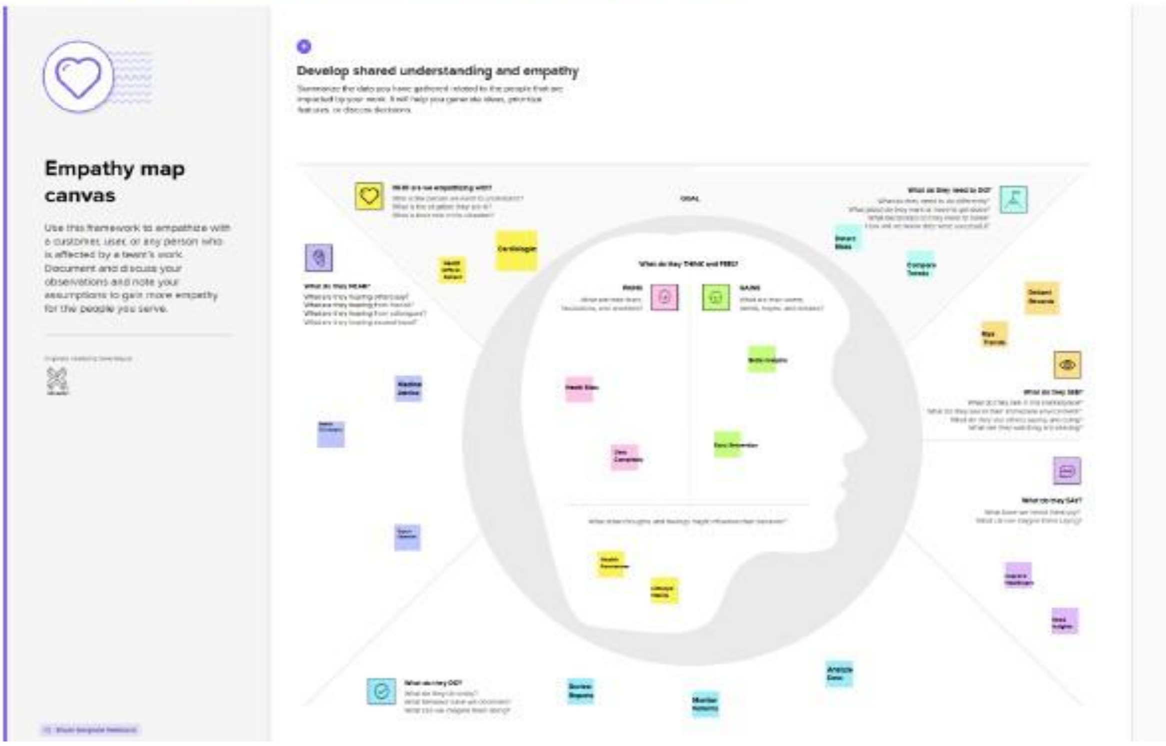
Date	5 July 2026
Team ID	
Project Name	**Heart Disease Analysis with Tableau**
Maximum Marks	4 Marks

Empathy Map Canvas:

An empathy map is a simple, easy-to-digest visual that captures knowledge about a user's behaviours and attitudes.

It is a useful tool to helps teams better understand their users.
Creating an effective solution requires understanding the true problem and the person who is experiencing it. The exercise of creating the map helps participants consider things from the user's perspective along with his or her goals and challenges.

Reference: https://www.mural.co/templates/empathy-map-canvas



2.3 Brainstorming

The brainstorming phase focused on generating innovative ideas and identifying effective approaches for analyzing heart disease data using interactive visualizations. Various healthcare challenges, user requirements, and analytical perspectives were explored to determine the most suitable solution. Techniques such as **How Might We (HMW)**, **Brainstorming Ideas**, **Affinity Mapping**, and **Solution Prioritization** were used to organize ideas and select the final project approach.

Figure 2.3: How Might We (HMW) Statements

Ideation Phase
Brainstorm & Idea Prioritization Template

Date	5 July 2026
Team ID	
Project Name	**Heart Disease Analysis with Tableau**
Maximum Marks	4 Marks

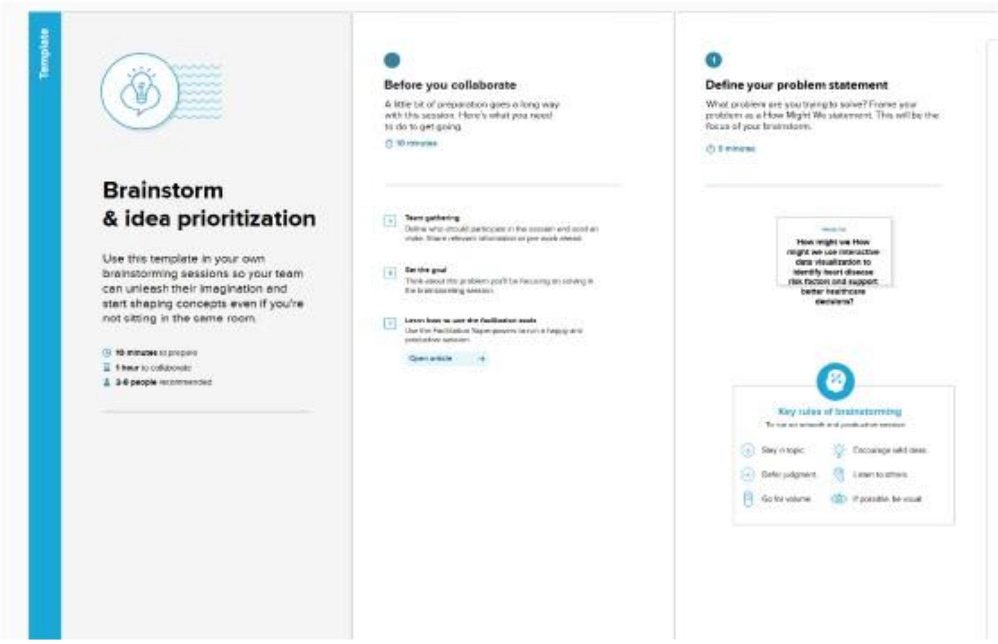
Brainstorm & Idea Prioritization Template:

Brainstorming provides a free and open environment that encourages everyone within a team to participate in the creative thinking process that leads to problem solving. Prioritizing volume over value, out-of-the-box ideas are welcome and built upon, and all participants are encouraged to collaborate, helping each other develop a rich amount of creative solutions.

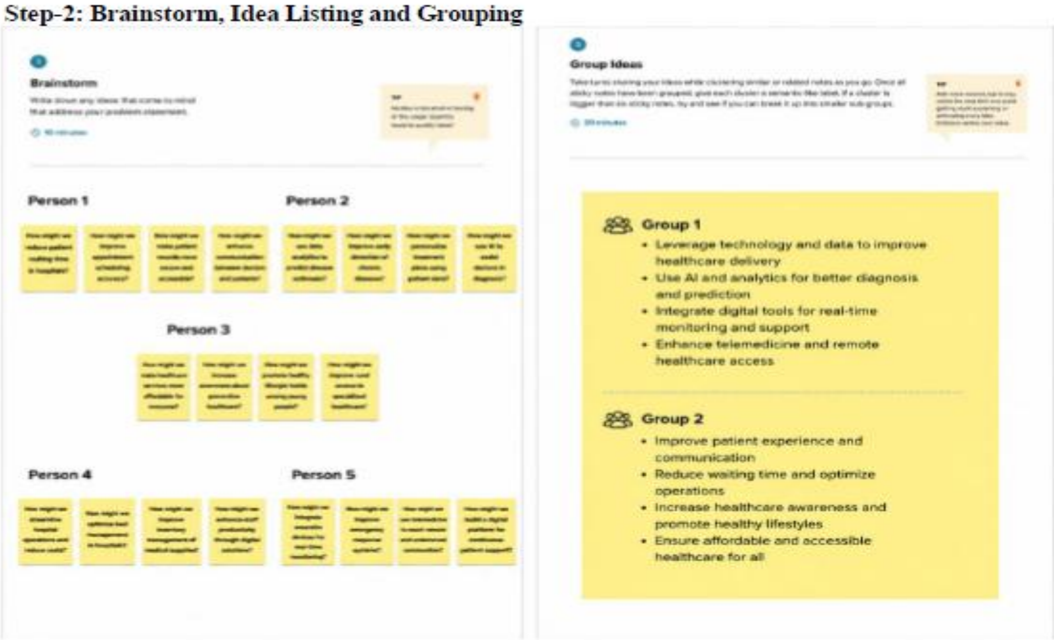
Use this template in your own brainstorming sessions so your team can unleash their imagination and start shaping concepts even if you're not sitting in the same room.

Reference: https://www.mural.co/templates/brainstorm-and-idea-prioritization

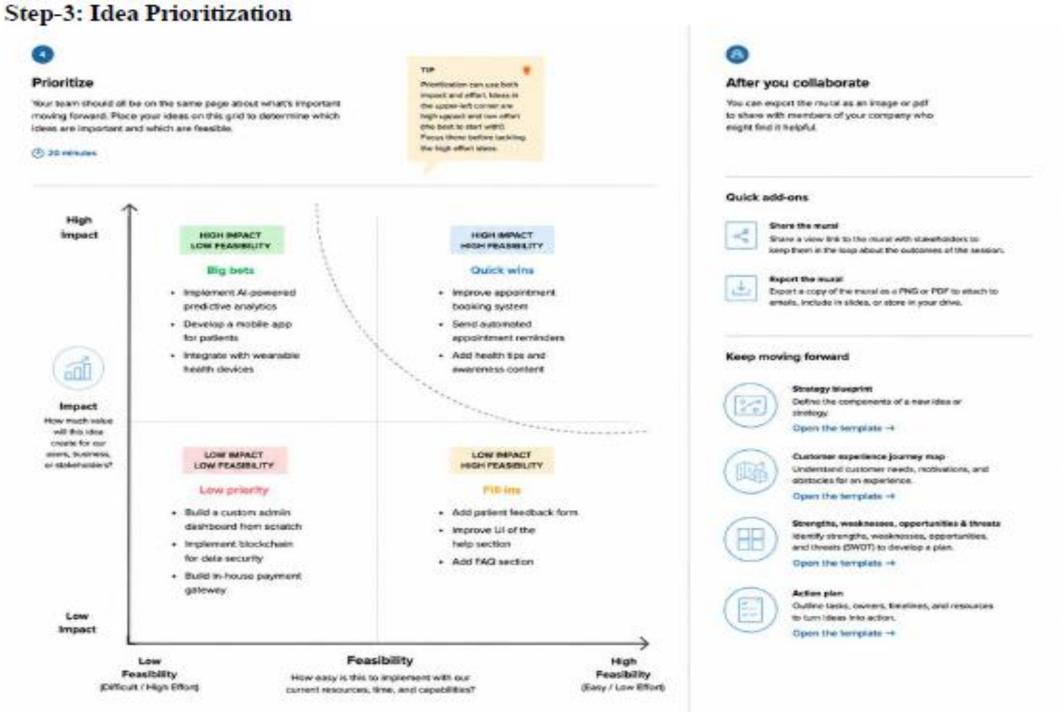
Step-1: Team Gathering, Collaboration and Select the Problem Statement



Step-2: Brainstorm, Idea Listing and Grouping



Step-3: Idea Prioritization



Conclusion

The brainstorming process helped identify the most practical and impactful solution for heart disease analysis. After evaluating multiple ideas, the team selected an interactive Tableau-based dashboard integrated with a Flask web application. This approach enables users to analyze demographic, lifestyle, and clinical factors through intuitive visualizations, supporting better understanding of heart disease patterns and informed healthcare decision-making.

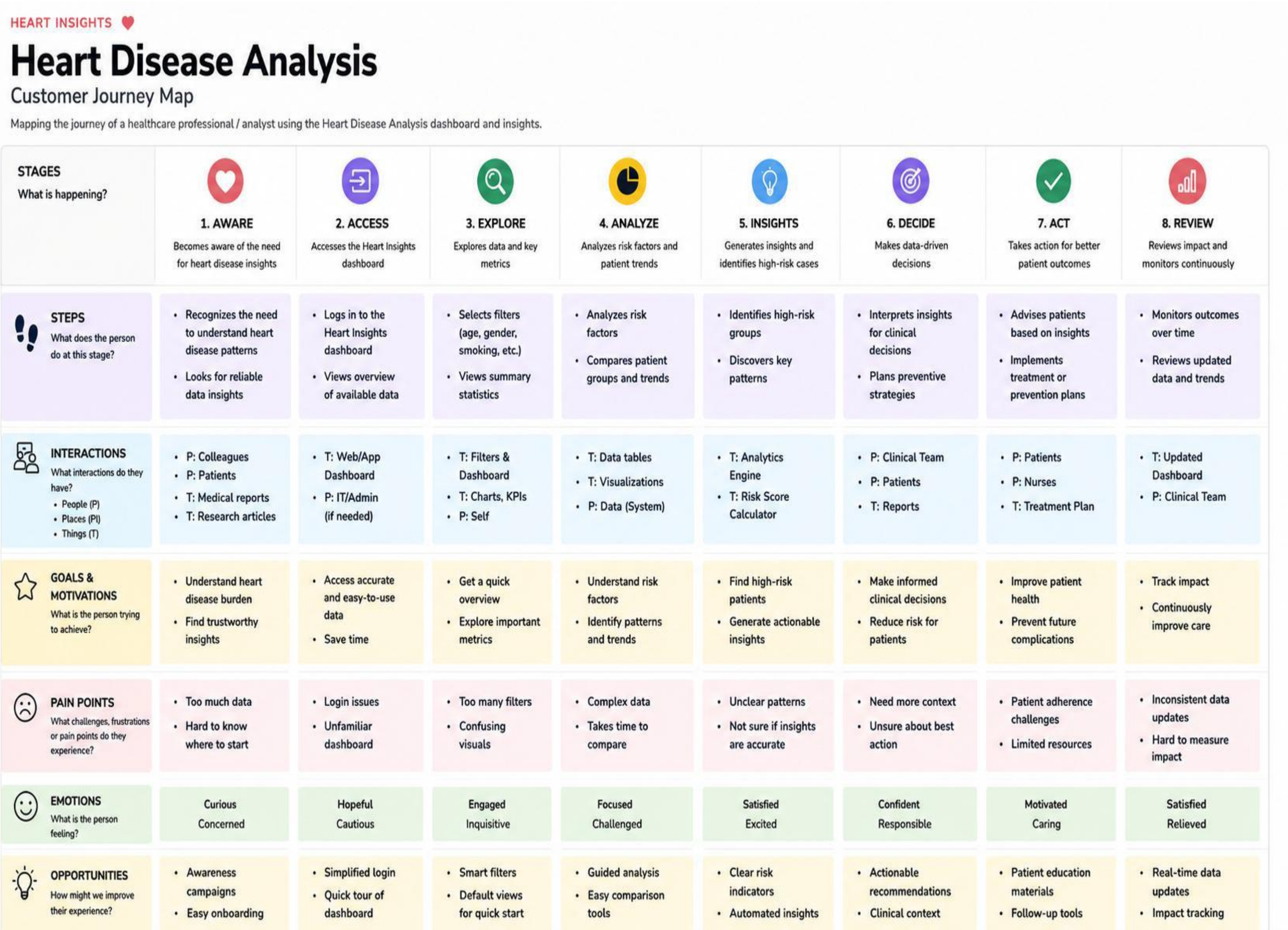
3. REQUIREMENT ANALYSIS

The Requirement Analysis phase focused on identifying stakeholder needs, defining the functional and non-functional requirements of the system, and designing the overall workflow for the Heart Disease Analysis project. This phase ensured that the proposed solution effectively addresses the problem statement while providing an intuitive, interactive, and user-friendly analytical platform for healthcare data visualization.

3.1 Customer Journey Map

The Customer Journey Map illustrates the step-by-step interaction of a user while exploring the Heart Disease Analysis platform. It highlights the user's objectives, touchpoints, challenges, and opportunities at each stage of the journey. This analysis helped in designing an intuitive dashboard and web application that simplifies healthcare data exploration and improves the overall user experience.

Figure 3.1: Customer Journey Map



HEART INSIGHTS

Heart Disease Analysis

Customer Journey Map

Mapping the journey of a healthcare professional / analyst using the Heart Disease Analysis dashboard and insights.

STAGES What is happening?	1. AWARE Becomes aware of the need for heart disease insights	2. ACCESS Accesses the Heart Insights dashboard	3. EXPLORE Explores data and key metrics	4. ANALYZE Analyzes risk factors and patient trends	5. INSIGHTS Generates insights and identifies high-risk cases	6. DECIDE Makes data-driven decisions	7. ACT Takes action for better patient outcomes	8. REVIEW Reviews impact and monitors continuously
STEPS What does the person do at this stage?	• Recognizes the need to understand heart disease patterns • Looks for reliable data insights	• Logs in to the Heart Insights dashboard • Views overview of available data	• Selects filters (age, gender, smoking, etc.) • Views summary statistics	• Analyzes risk factors • Compares patient groups and trends	• Identifies high-risk groups • Discovers key patterns	• Interprets insights for clinical decisions • Plans preventive strategies	• Advises patients based on insights • Implements treatment or prevention plans	• Monitors outcomes over time • Reviews updated data and trends
INTERACTIONS What interactions do they have? • People (P) • Places (Pl) • Things (T)	• P: Colleagues • P: Patients • T: Medical reports • T: Research articles	• T: Web/App Dashboard • P: IT/Admin (if needed)	• T: Filters & Dashboard • T: Charts, KPIs • P: Self	• T: Data tables • T: Visualizations • P: Data (System)	• T: Analytics Engine • T: Risk Score Calculator	• P: Clinical Team • P: Patients • T: Reports	• P: Patients • P: Nurses • T: Treatment Plan	• T: Updated Dashboard • P: Clinical Team
GOALS & MOTIVATIONS What is the person trying to achieve?	• Understand heart disease burden • Find trustworthy insights	• Access accurate and easy-to-use data • Save time	• Get a quick overview • Explore important metrics	• Understand risk factors • Identify patterns and trends	• Find high-risk patients • Generate actionable insights	• Make informed clinical decisions • Reduce risk for patients	• Improve patient health • Prevent future complications	• Track impact • Continuously improve care
PAIN POINTS What challenges, frustrations or pain points do they experience?	• Too much data • Hard to know where to start	• Login issues • Unfamiliar dashboard	• Too many filters • Confusing visuals	• Complex data • Takes time to compare	• Unclear patterns • Not sure if insights are accurate	• Need more context • Unsure about best action	• Patient adherence challenges • Limited resources	• Inconsistent data updates • Hard to measure impact
EMOTIONS What is the person feeling?	Curious Concerned	Hopeful Cautious	Engaged Inquisitive	Focused Challenged	Satisfied Excited	Confident Responsible	Motivated Caring	Satisfied Relieved
OPPORTUNITIES How might we improve their experience?	• Awareness campaigns • Easy onboarding	• Simplified login • Quick tour of dashboard	• Smart filters • Default views for quick start	• Guided analysis • Easy comparison tools	• Clear risk indicators • Automated insights	• Actionable recommendations • Clinical context	• Patient education materials • Follow-up tools	• Real-time data updates • Impact tracking

The customer journey demonstrates how users progress from understanding the purpose of the application to exploring dashboards, analyzing healthcare insights, interacting with filters, and interpreting Tableau Story findings. Identifying user pain points during this process enabled the development of a solution that provide clear visualizations, interactive analytics, and easy navigation for healthcare professionals, researchers, and patients.

3.2 SOLUTION REQUIREMENT

The Heart Disease Analysis project was designed to transform raw healthcare data into meaningful visual insights through interactive dashboards and a web-based interface. The solution requirements were identified to ensure that the application provides effective data visualization, user-friendly interaction, and reliable analytical capabilities for healthcare professionals, researchers, and patients.

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	5 July 2026
Team ID	
Project Name	**Heart Disease Analysis with Tableau**
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

Functional Requirements

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Dataset Management	Import the Heart Disease dataset into Tableau. Validate and organize the dataset. Prepare data for visualization.
FR-2	Data Preprocessing	Handle missing or inconsistent data. Create calculated fields and measures. Assign appropriate data types.
FR-3	Dashboard Analysis	Develop interactive Tableau dashboards. Display KPI cards, charts, and graphs. Support interactive filtering and data exploration.
FR-4	Tableau Story	Create a Tableau Story using dashboard insights. Present analytical findings sequentially. Enable navigation through story points.
FR-5	Data Visualization	Visualize demographic, lifestyle, and health-related trends. Compare different patient groups. Highlight heart disease risk factors.
FR-6	Insight Generation	Generate meaningful healthcare insights from dashboard analysis. Support decision-making through interactive visualizations.

Non-Functional Requirements

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The Tableau dashboards shall provide an intuitive and user-friendly interface for data exploration.
NFR-2	Accuracy	Visualizations, KPI cards, and calculated fields shall accurately represent the underlying dataset.
NFR-3	Reliability	Dashboards and Tableau Story shall consistently display correct information without errors.
NFR-4	Performance	Dashboards, filters, and visualizations shall respond efficiently with minimal loading time.
NFR-5	Availability	Tableau dashboards and stories shall be accessible whenever the Tableau platform is available.
NFR-6	Scalability	The project shall support additional datasets, dashboards, visualizations, and analytical features in future enhancements.

3.3 DATA FLOW DIAGRAM

The Data Flow Diagram (DFD) illustrates the movement of data throughout the Heart Disease Analysis system. It represents how healthcare data is collected, processed, visualized, and presented to users through interactive dashboards and a web-based interface. The DFD provides a clear understanding of the relationship between the dataset, Tableau analytics, Flask application, and end users.

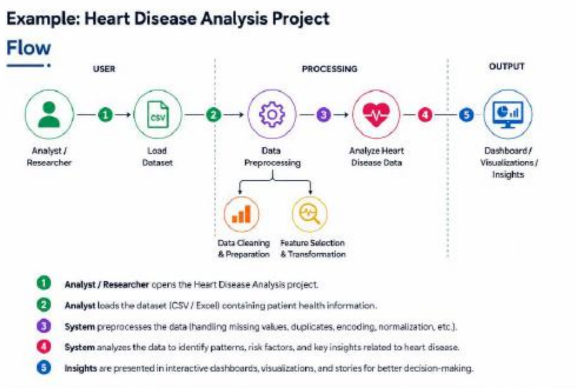
Figure 3.2: Data Flow Diagram

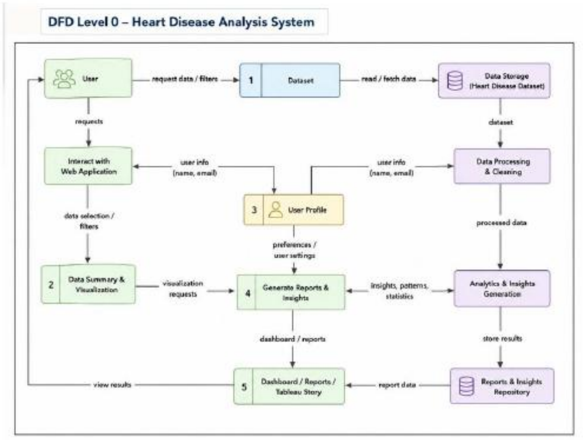
Project Design Phase-II
Data Flow Diagram & User Stories

Date	7 July 2026
Team ID	
Project Name	**Heart Disease Analysis with Tableau**
Maximum Marks	4 Marks

Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.





User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Data Analyst	Dataset Import	USN-1	As a data analyst, I want to import the heart disease dataset into Tableau for analysis.	Dataset is imported successfully.	High	Sprint-1
Data Analyst	Data Preparation	USN-2	As a data analyst, I want to preprocess the dataset before visualization.	Data is cleaned and ready for analysis.	High	Sprint-1
Data Analyst	Dashboard Development	USN-3	As a data analyst, I want to create interactive dashboards for heart disease analysis.	Dashboards display correct visualizations.	High	Sprint-1
Healthcare Professional	Risk Analysis	USN-4	As a healthcare professional, I want to identify high-risk patient groups.	Dashboard highlights risk factors correctly.	High	Sprint-1
Healthcare Professional	Interactive Filters	USN-5	As a healthcare professional, I want to filter patient data by demographic and lifestyle factors.	Filters update all visualizations correctly.	High	Sprint-2
Researcher	Trend Analysis	USN-6	As a researcher, I want to compare disease trends across different patient groups.	Charts update dynamically based on selected filters.	Medium	Sprint-2
Researcher	Visualization Analysis	USN-7	As a researcher, I want to analyze relationships between different health indicators.	Visualizations display meaningful comparisons.	Medium	Sprint-2
Public Health Officer	Healthcare Insights	USN-8	As a public health officer, I want to identify disease trends for awareness planning.	Dashboard presents actionable insights.	High	Sprint-2
Public Health Officer	Tableau Story	USN-9	As a public health officer, I want to review the Tableau Story for summarized findings.	Story displays all analytical insights correctly.	Medium	Sprint-2
Decision Maker	KPI Monitoring	USN-10	As a decision maker, I want to monitor key healthcare indicators using KPI cards.	KPI cards display accurate values.	High	Sprint-3

3.4 TECHNOLOGY STACK

The Heart Disease Analysis project utilizes a combination of business intelligence, web development, and visualization technologies to transform raw healthcare data into interactive analytical dashboards. Tableau is used for data visualization and storytelling, while Flask provides a lightweight web framework to integrate the dashboards into a user-friendly application. The selected technologies ensure efficient data analysis, accessibility, and an enhanced user experience.

Table 3.1: Technology Stack

Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	7 July 2026
Team ID	
Project Name	**Heart Disease Analysis with Tableau**
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Order processing during pandemics for offline mode

Reference: https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/

Heart Disease Analysis with Tableau:

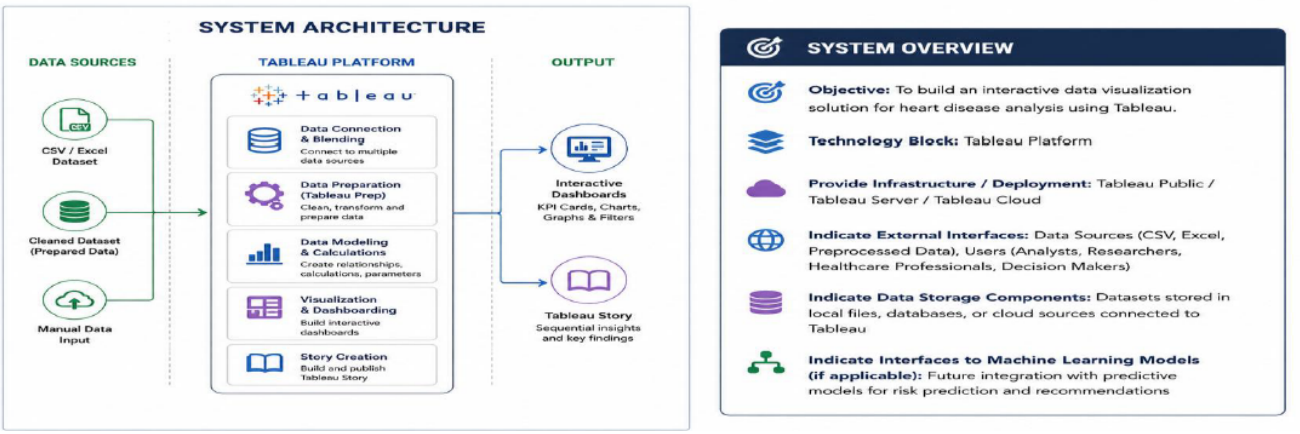


Table-1: Components & Technologies

S.No	Component	Description	Technology
1	Data Source	Heart disease dataset containing patient demographics, lifestyle, and clinical information.	CSV Dataset
2	Data Preparation	Cleaning, formatting, and preprocessing of the dataset before visualization.	Tableau Desktop
3	Calculated Fields	Creation of calculated fields, measures, and KPIs for analysis.	Tableau Calculated Fields
4	Data Visualization	Development of interactive charts, graphs, and visual analytics.	Tableau Desktop
5	Dashboard Development	Creation of interactive dashboards with filters and KPI cards.	Tableau Dashboard
6	Story Development	Presentation of analytical insights using sequential story points.	Tableau Story
7	Dashboard Publishing	Publishing dashboards and stories for online access and sharing.	Tableau Public
8	Interactive Filters	Enable users to filter data by demographic and health-related attributes.	Tableau Filters
9	KPI Cards	Display key healthcare metrics for quick analysis and decision-making.	Tableau Dashboard
10	File Storage	Stores datasets and project resources locally.	Local File System
11	Infrastructure	Local development and dashboard publishing environment.	Tableau Desktop & Tableau Public

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Tools	Tools used for data visualization and dashboard development.	Tableau Public
2	Data Accuracy	Ensures accurate calculations, KPI values, and visualizations.	Tableau Calculated Fields
3	Scalable Architecture	Dashboards can be extended with additional datasets, charts, and stories.	Tableau Dashboard
4	Availability	Published dashboards and stories are accessible online through Tableau Public.	Tableau Public
5	Performance	Optimized dashboards with interactive filters and efficient visualization rendering.	Tableau Desktop & Tableau Public

Technology Description

The project combines Tableau Desktop for interactive dashboard development and Tableau Public for publishing dashboards and stories. Flask serves as the backend framework, enabling seamless integration of Tableau visualizations into a web application. HTML, CSS, Bootstrap, and JavaScript are used to build a responsive and user-friendly interface, while Python handles backend functionality. GitHub is utilized for version control and project management, and Visual Studio Code serves as the primary development environment.

4. PROJECT DESIGN

The Project Design phase focuses on transforming the identified requirements into a practical and effective solution. Based on the problem statement and stakeholder needs, an interactive Tableau-based analytics platform was designed to simplify healthcare data analysis through dashboards, visualizations, KPI cards, and web integration. This phase defines how the proposed solution addresses the challenges identified during the ideation and requirement analysis stages.

4.1 PROBLEM SOLUTION FIT

The Heart Disease Analysis project addresses the challenge of interpreting complex healthcare datasets by providing an interactive business intelligence solution using Tableau. Traditional methods of analyzing patient records often involve manual interpretation of large datasets, making it difficult to identify meaningful trends and relationships. The proposed solution transforms raw healthcare data into interactive dashboards that enable users to analyze demographic, lifestyle, and clinical factors efficiently.

The solution incorporates **three dashboards**, **ten interactive visualizations**, **KPI cards**, **calculated fields**, **interactive filters**, and **Tableau Story** to present healthcare information in a structured and user-friendly manner. By integrating these visualizations into a Flask-based web application, the project improves accessibility and enables healthcare professionals, researchers, and patients to explore analytical insights through an intuitive interface.

Figure 4.1: Problem Solution Fit

Project Design Phase
Problem – Solution Fit Template

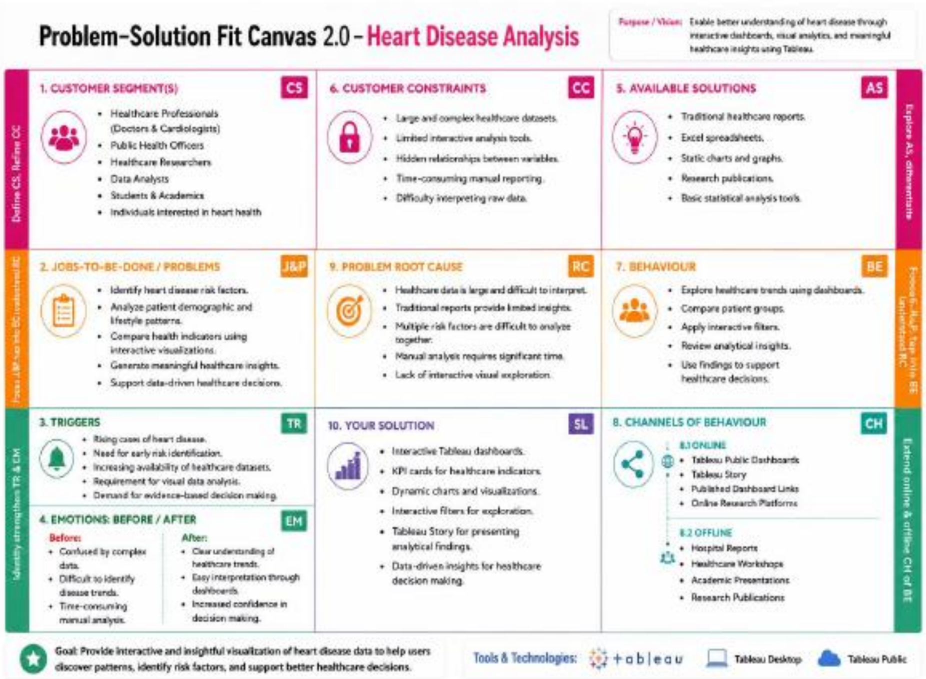
Date	7 July 2026
Team ID	
Project Name	**Heart Disease Analysis with Tableau**
Maximum Marks	2 Marks

Problem – Solution Fit Template:

The Problem-Solution Fit simply means that you have found a problem with your customer and that the solution you have realized for it actually solves the customer's problem. It helps entrepreneurs, marketers and corporate innovators identify behavioral patterns and recognize what would work and why

Purpose:

- ❑ Solve complex problems in a way that fits the state of your customers.
- ❑ Succeed faster and increase your solution adoption by tapping into existing mediums and channels of behavior.
- ❑ Sharpen your communication and marketing strategy with the right triggers and messaging.
- ❑ Increase touch-points with your company by finding the right problem-behavior fit and building trust by solving frequent annoyances, or urgent or costly problems.
- ❑ **Understand the existing situation in order to improve it for your target group.**



The proposed solution successfully aligns with the identified problem by converting healthcare data into meaningful visual insights that support trend analysis, risk identification, and informed decision-making. The combination of interactive dashboards and web integration ensures that users can easily explore healthcare information without requiring advanced technical knowledge.

4.2 PROPOSED SOLUTION

The proposed solution is an interactive **Heart Disease Analysis platform** developed using Tableau and integrated into a Flask web application. The solution consolidates patient demographic information, lifestyle factors, and medical conditions into a centralized analytical environment where users can explore healthcare trends through interactive dashboards and visualizations The platform consists of **three analytical dashboards**, **ten visualizations**, **KPI cards**, **interactive filters**, and a **Tableau Story** that collectively provide a comprehensive understanding of heart disease patterns. Users can compare patient groups, analyze relationships between health indicators, and explore risk factors such as age, BMI, smoking habits, diabetes, stroke history, and physical activity.

Project Design Phase
Proposed Solution Template

Date	7 July 2026
Team ID	
Project Name	**Heart Disease Analysis with Tableau**
Maximum Marks	2 Marks

Proposed Solution Template

S.No.	Parameter	Description
1	**Problem Statement (Problem to be Solved)**	Heart disease is one of the leading causes of mortality worldwide. Analyzing large healthcare datasets to identify disease patterns and risk factors can be challenging using traditional reports. There is a need for an interactive data visualization solution that transforms raw healthcare data into meaningful insights to support better understanding and informed healthcare decision-making.
2	**Idea / Solution Description**	Develop a Heart Disease Analysis solution using **Tableau** to create interactive dashboards, KPI cards, charts, filters, and Tableau Stories. The solution enables users to explore healthcare trends, demographic patterns, lifestyle risk factors, and analytical insights through interactive visualizations.
3	**Novelty / Uniqueness**	The solution leverages Tableau's interactive visualization and storytelling capabilities to simplify complex healthcare data. By combining dashboards, KPI cards, filters, and Tableau Stories, users can explore heart disease trends in an intuitive and visually engaging manner.
4	**Social Impact / Customer Satisfaction**	The solution increases awareness of heart disease risk factors and supports healthcare professionals, researchers, students, and public health organizations in understanding disease trends. Interactive visualizations promote data-driven healthcare decisions and contribute to preventive healthcare awareness.
5	**Business Model (Revenue Model)**	The solution can be offered as a healthcare analytics dashboard for hospitals, clinics, research organizations, educational institutions, and public health agencies. Revenue opportunities include institutional subscriptions, customized analytical dashboards, consulting services, and training programs on healthcare data visualization.
6	**Scalability of the Solution**	The solution is designed to support additional healthcare datasets, new dashboards, advanced visualizations, predictive analytics integration, and future expansion using Tableau's scalable analytics platform, making it adaptable to evolving healthcare requirements.

To improve accessibility, the Tableau dashboards and story are embedded within a Flask-based web application that provides dedicated pages for project information, dashboards, insights, technologies, developer details, and contact information. This integration enables users to access healthcare analytics through a centralized and user-friendly platform.

4.3 SOLUTION ARCHITECTURE

The Solution Architecture illustrates the overall structure of the Heart Disease Analysis system and the interaction between its major components. The architecture begins with the Heart Disease dataset, which serves as the primary data source. The dataset is processed in Tableau Desktop, where interactive dashboards, KPI cards, calculated fields, and Tableau Story are developed.

The published dashboards are hosted on Tableau Public and integrated into the Flask web application. Users access the application through a web browser, where they can interact with dashboards, apply filters, explore stories, and analyze healthcare insights through an intuitive interface. This architecture ensures efficient data visualization, seamless integration, and improved accessibility while maintaining a simple and scalable system design**.**

Figure 4.2: Solution Architecture

Project Design Phase
Solution Architecture

Date	7 July 2026
Team ID	
Project Name	Heart Disease Analysis with Tableau
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Solution Architecture Diagram:

Figure 1: Architecture and data flow of the Heart Disease Analysis application

The architecture demonstrates a clear flow of information from the healthcare dataset to the end user. By combining Tableau for visualization and Flask for web integration, the solution provides a lightweight yet effective platform for healthcare analytics and interactive decision support.

5. PROJECT PLANNING & SCHEDULING

The Project Planning and Scheduling phase outlines the systematic execution of the Heart Disease Analysis project. The development process was divided into multiple sprints, allowing the team to complete data preparation, dashboard development, web integration, testing, and documentation in an organized manner. Sprint-based planning ensured effective task management, timely completion, and continuous progress throughout the project lifecycle.

5.1 Project Planning

The project was planned using an iterative sprint-based approach to ensure the systematic completion of all development activities. Each sprint focused on a specific set of objectives, including data preprocessing, dashboard development, KPI creation, Tableau Story design, Flask web integration, testing, and documentation. This structured planning approach improved project organization, enabled continuous monitoring of progress, and ensured timely completion of all deliverables.

Table 5.1: Product Backlog & Sprint Planning

Project Planning Phase

Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)

Date	8 July 2026
Team ID	
Project Name	**Heart Disease Analysis with Tableau**
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule & Estimation

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Data Collection	USN-1	Collect the heart disease dataset for analysis.	3	High	Team
Sprint-1	Data Preparation	USN-2	Clean, preprocess, and organize the dataset.	3	High	Team
Sprint-1	Data Modeling	USN-3	Create calculated fields, measures, and KPIs.	5	High	Team
Sprint-2	Data Visualization	USN-4	Create interactive charts and graphs using Tableau.	5	High	Team
Sprint-2	Dashboard Development	USN-5	Develop interactive dashboards with filters and KPI cards.	8	High	Team
Sprint-2	Story Development	USN-6	Create Tableau Story to present analytical insights.	5	High	Team
Sprint-3	Dashboard Publishing	USN-7	Publish dashboards and stories to Tableau Public.	3	Medium	Team

Table 5.3: Velocity & Burndown

Sprint-3	Dashboard Validation	USN-8	Validate dashboards, filters, KPIs, and story navigation.	5	High	Team
Sprint-4	Documentation	USN-9	Prepare project documentation, reports, and diagrams.	5	High	Team
Sprint-4	Testing & Final Review	USN-10	Perform testing, validate outputs, and finalize the project.	5	High	Team

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed	Sprint Release Date (Actual)
Sprint-1	11	2 Days	01 Jul 2026	02 Jul 2026	11	02 Jul 2026
Sprint-2	18	2 Days	03 Jul 2026	04 Jul 2026	18	04 Jul 2026
Sprint-3	8	1 Day	05 Jul 2026	05 Jul 2026	8	05 Jul 2026
Sprint-4	10	2 Days	06 Jul 2026	07 Jul 2026	10	07 Jul 2026

Story Point Distribution

- Sprint-1: 3 + 3 + 5 = 11
- Sprint-2: 5 + 8 + 5 = 18
- Sprint-3: 3 + 5 = 8
- Sprint-4: 5 + 5 = 10

Total Story Points = 47

Velocity:
Imagine we have a 10-day sprint duration, and the velocity of the team is 20 (points per sprint). Let's calculate the team's average velocity (AV) per iteration unit (story points per day)

$$AV = \frac{sprint\ duration}{velocity} = \frac{20}{10} = 2$$

Burndown Chart:

A burn down chart is a graphical representation of work left to do versus time. It is often used in agile software development methodologies such as Scrum. However, burn down charts can be applied to any project containing measurable progress over time.

Day	Remaining Story Points
Day 1	20
Day 2	18
Day 3	15
Day 4	12
Day 5	9
Day 6	6
Day 7	4
Day 8	2
Day 9	1
Day 10	0

Project Planning Summary

The project was executed through multiple development sprints, beginning with dataset preparation and requirement analysis, followed by dashboard design, visualization development, calculated field implementation, web integration, testing, and final documentation. The sprint-based methodology enabled efficient task allocation, continuous progress tracking, and successful completion of the project within the planned schedule.

6. FUNCTIONAL AND PERFORMANCE TESTING

The Functional and Performance Testing phase was conducted to verify that the Heart Disease Analysis application performs as expected under normal operating conditions. The testing process evaluated the functionality of the Tableau dashboards, web application, interactive filters, KPI cards, and navigation features to ensure accuracy, usability, and reliability. The results confirmed that the application successfully met the intended project requirements.

6.1 Functional Testing

Functional testing was performed to validate the core features of the Heart Disease Analysis application. Each module was tested to ensure that it functioned correctly and produced the expected results. The testing included dashboard loading, navigation, KPI display, filter interaction, Tableau Story accessibility, and web page responsiveness.

Test Case	Expected Result	Actual Result	Status
Dataset Loading	**Dataset loads successfully into Tableau**	**Loaded Successfully**	**Pass**
Data Preprocessing	**Dataset is cleaned and prepared correctly**	**Completed Successfully**	**Pass**
Dashboard 1 Loading	**Dashboard 1 displays all visualizations correctly**	**Loaded Successfully**	**Pass**
Dashboard 2 Loading	**Dashboard 2 displays all visualizations correctly**	**Loaded Successfully**	**Pass**
KPI Cards Display	**KPI cards display accurate values**	**Displayed Successfully**	**Pass**
Interactive Filters	**Filters update charts and dashboards correctly**	**Working Successfully**	**Pass**
Tableau Story	**Story loads with all story points**	**Loaded Successfully**	**Pass**
Dashboard Publishing	**Dashboards and Story accessible on Tableau Public**	**Published Successfully**	**Pass**

Functional Testing Summary

The functional testing results indicate that all major features of the application operate as expected. The dashboards, stories, filters, KPI cards, navigation components, and web pages function correctly without any observed issues.

6.2 Performance Testing

Performance testing was conducted to evaluate the responsiveness and usability of the application during normal usage. The assessment focused on dashboard loading, filter responsiveness, visualization rendering, and overall application performance within the Flask web environment.

Table 6.2: Performance Testing

Project Performance Test

Date	8 July 2026
Team ID	
Project Name	**Heart Disease Analysis with Tableau**
Maximum Marks	

Model Performance Testing:

S.No.	Parameter	Screenshot / Values
1	Data Rendered	**4,500 Records Loaded**
2	Data Preprocessing	**Data Cleaning, Formatting & Validation Completed**
3	Utilization of Filters	**7 Interactive Filters Applied**
4	Calculation Fields Used	**3 Calculated Fields Created**
5	Dashboard Design	**2 Interactive Dashboards, 10 Visualizations, 7 KPI Cards**
6	Story Design	**1 Tableau Story with 6 Story Points**

Performance Testing Summary

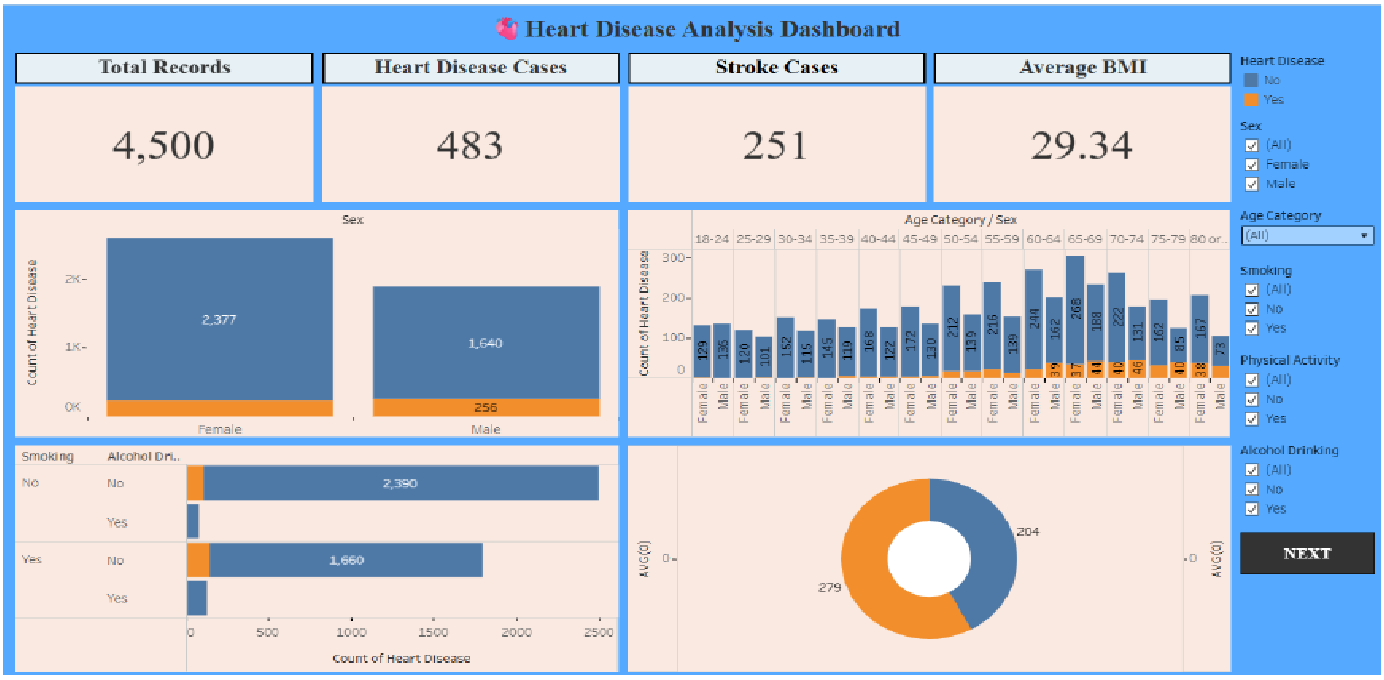
The application demonstrated stable performance during testing. Dashboards and stories loaded successfully, filters responded accurately, and all visual components were rendered correctly. The Flask web application provided a smooth and responsive user experience throughout the testing process.

7. DOCUMENTATION & DEMO

7.1 Final Dashboard

The final Tableau dashboard presents a comprehensive analysis of heart disease by combining demographic, lifestyle, and clinical indicators into an interactive visualization platform. The dashboard enables users to explore healthcare trends, compare patient groups, and identify major risk factors through KPI cards, interactive charts, calculated fields, and dynamic filters. The centralized dashboard provides meaningful insights that support healthcare awareness and informed decision-making.

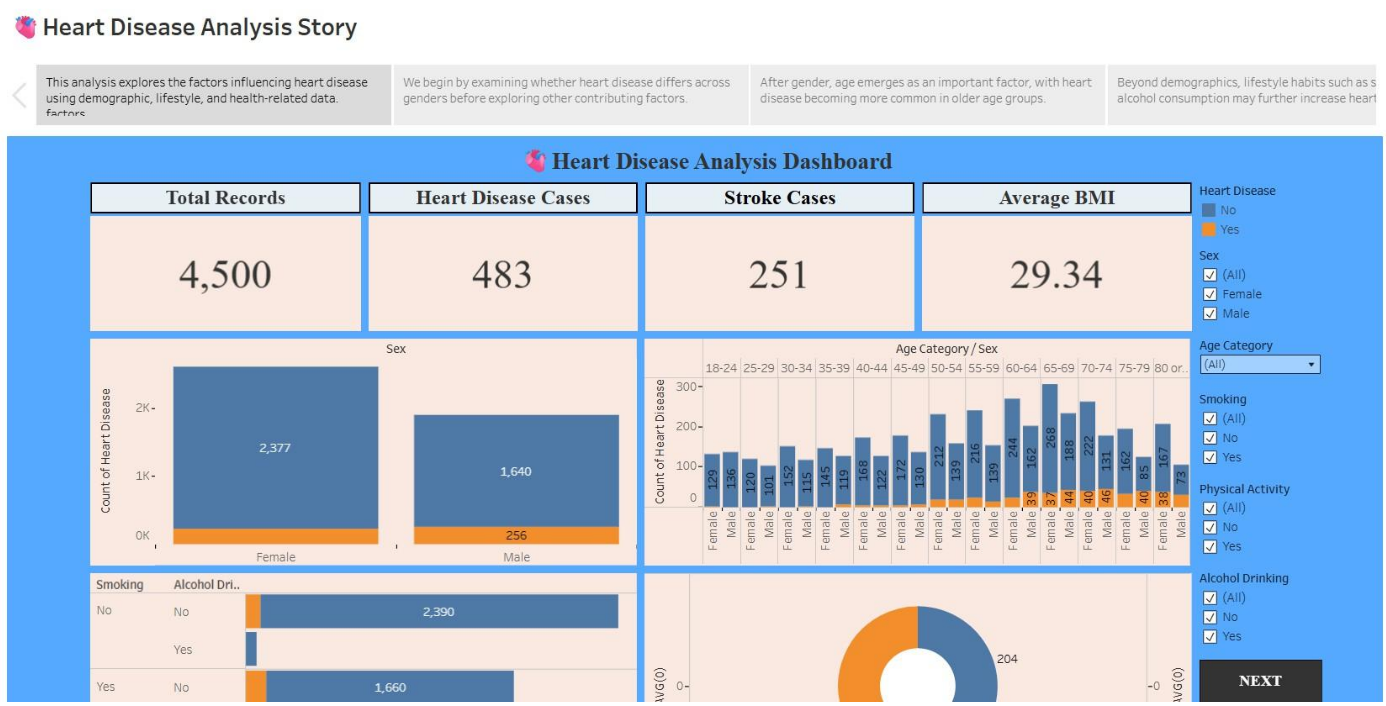
Figure 7.1: Final Tableau Dashboard



7.2 Tableau Story

The Tableau Story presents the analytical findings in a structured sequence, guiding users through important healthcare insights derived from the dataset. Each story point focuses on a different aspect of heart disease analysis, enabling users to understand demographic patterns, lifestyle influences, medical conditions, and overall healthcare trends through an intuitive storytelling approach.

Figure 7.2: Tableau Story



7.3 GitHub Repository

The complete project source code, website resources, and supporting files are maintained using GitHub for version control and collaborative development. The repository ensures organized project management, simplified maintenance, and easy access to the implementation files.

GITHUB REPOSITORY

Web Application Repository URL:

https://github.com/thiragatisuryasri-analyst/Heart-Disease-Analysis-with-Tableau

7.5 Tableau Public Links

Dashboard:

https://public.tableau.com/views/HeartDiseaseAnalysis-1_178308620187G0/Home?:language=en-USs:sid=s:redirect=auths:display_count=ns:origin=viz_share_link

Tableau Story:

https://public.tableau.com/views/HeartDiseaseAnalysis-2_17832385228620/Story?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

7.6 Project Summary

The Heart Disease Analysis project successfully transformed raw healthcare data into an interactive analytical platform using Tableau and Flask. Through three dashboards, ten visualizations, KPI cards, calculated fields, interactive filters, and Tableau Story, the project enables users to analyze heart disease trends, identify risk factors, and explore healthcare insights efficiently. The integration of Tableau with a Flask web application enhances accessibility and provides a centralized platform for healthcare analytics, supporting data-driven awareness and decision-making.

8. ADVANTAGES

Advantages:

- Interactive dashboards simplify the analysis of complex healthcare data.
- KPI cards provide a quick overview of important healthcare indicators.
- Interactive filters enable users to explore specific patient groups dynamically.
- Tableau Story presents analytical findings in a structured and easy-to-understand format.

9. CONCLUSION

The **Heart Disease Analysis using Tableau** project successfully transformed raw healthcare data into an interactive analytical platform that supports meaningful exploration of heart disease trends and associated risk factors. By utilizing Tableau dashboards, KPI cards, calculated fields, interactive filters, and Tableau Story, the project enables users to analyze demographic, lifestyle, and clinical indicators through intuitive visualizations.

10. FUTURE SCOPE

The current project can be enhanced further by incorporating advanced healthcare analytics and additional interactive features. Future improvements may include:

- Integration of real-time healthcare datasets for continuous analysis.
- Development of predictive models for early heart disease risk assessment.
- Addition of personalized health recommendations based on patient profiles.
- Deployment of the application on cloud platforms for wider accessibility.
- Development of mobile-responsive and cross-platform versions of the application.
- Integration with hospital information systems and electronic health records.
- Expansion of analytical dashboards using larger and more diverse healthcare datasets.

11. APPENDIX

Tableau Dashboard
https://public.tableau.com/views/HeartDiseaseAnal
ysis-1_178308620187G0/Home?:language=en-
USs:sid=s:redirect=auths:display_count=ns:origin=viz_share_link

Tableau Story
https://public.tableau.com/views/HeartDiseaseAnalysis-
2_17832385228620/Story?:language=en-
US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

GitHub Repository
https://github.com/thiragatisuryasri-analyst/Heart-Disease-Analysis-with-Tableau.git

Project Demo
A demonstration of the project showcasing data preprocessing, interactive dashboards, KPI cards, filters, and Tableau Story presentation.
Development Tools

- Tableau Desktop
- Tableau Public
- Microsoft Excel (Data Preparation)
- GitHub